
Pablo Arancibia

Sr. Software Engineer

Contact info: pabloarancibia.dw@gmail.com
Portfolio: <https://github.com/pabloarancibia>
Experience: 10+ years.

www.linkedin.com/in/arancibiapablo
Location: Málaga, Spain.

Profile

Results-oriented Senior Python Engineer with over 10 years of experience architecting high-performance, cloud-native backend systems. Specializes in building scalable Microservices using **FastAPI, gRPC, and Hexagonal Architecture** within the Fintech sector. Proven track record of delivering compliant (PSD2/SCA), high-availability solutions for handling sensitive financial data.

Combines deep roots in low-level engineering (Air Force R&D) with modern expertise in **Kubernetes, Terraform, and Event-Driven Design**. Passionate about software craftsmanship, **TDD**, and engineering standards. Currently leveraging this backend expertise to engineer production-grade infrastructure for **Generative AI and RAG** applications..

Core Competencies

Languages: Python (Expert - 10+ years), TypeScript, JavaScript, SQL, C++.

Frameworks: FastAPI, Django, Pytest, Node.js. - Angular

Cloud & Infrastructure:

- **Professional:** AWS & Google Cloud Platform (Cloud Run, Functions, Pub/Sub, IAM).
- **Academic/R&D:** Microsoft Azure (Azure OpenAI, Azure AI Search, Semantic Kernel).
- **IaC:** Terraform (Multi-cloud provisioning), Docker, Kubernetes (AKS/EKS concepts).
- **Architecture:** Microservices, Event-Driven Architecture (EDA), Hexagonal (Ports & Adapters), Domain-Driven Design (DDD), Serverless.
- **Data & AI Engineering:** PostgreSQL, Redis, RabbitMQ, **Vector Databases** (ChromaDB, Azure AI Search), **RAG** Systems, **MCP** (Model Context Protocol), **NoSql** (MongoDB, DynamoDB, Firestore).
- **Fintech & Security:** PSD2, Open Banking, OAuth2/OIDC, JWT, Financial Data Compliance.

Professional Experience

ATM | Spain (Remote) Senior Software Engineer | Jan 2026 –

Aug 2026 Led the end-to-end implementation of a new backend architecture from scratch, establishing a scalable and robust software foundation.

- **Technology Stack:** Built highly reliable core services utilizing Django, Django REST Framework (DRF), Sentry, Kubernetes, and Redis.
- **Strategic Handoff:** Executed a temporary, high-impact role focused strictly on architecture and software design, successfully delivering a maintainable system for the company's permanent engineering staff.

Brite Payments | Spain (Remote) Senior Software Engineer |

May 2024 – Dec 2025 - Architecting secure backend microservices and data pipelines using Python (FastAPI) and Google Cloud Platform.

- **Open Banking Implementation:** Developed robust backend solutions strictly adhering to **PSD2 regulations** and standards (BerlinGroup, Obie), handling sensitive KYC, Account Information (AIS), and Payment Initiation (PIS) data.
- **High-Availability Architecture:** Optimized **GCP infrastructure** (Cloud Run, Cloud Functions) to handle high-volume request spikes, implementing retry logic and circuit breakers to ensure 99.9% uptime.
- **Security & Identity:** Implemented advanced authentication flows using **OAuth2 and JWT**, managing secure data exchange between internal microservices and external providers.
- **Event-Driven Systems:** Built scalable data processing pipelines to ensure data consistency and synchronization across distributed databases, prioritizing low latency and data integrity.

Freelance | Spain (Remote) Senior Software Developer | 2024

- **QA Automation:** Architected a comprehensive Testing Suite Framework using **Pytest** and Django for a private client, significantly reducing regression bugs.

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- **CI/CD Implementation:** Built and deployed CI/CD pipelines via Bitbucket to automate quality assurance, testing, and deployment processes.
 - **Legacy Migration:** Led the integration of legacy Node.js services with modern Python backend logic, improving system maintainability and performance.

ECOM Chaco | Argentina (Remote) Lead Fullstack Developer

| 2020 – 2023

- **Agile Leadership:** Led development teams using Scrum methodologies; managed requirements analysis, task prioritization, and architectural decision-making.
- **Scalable Architecture:** Designed and optimized scalable architectures using **Python (Django/DRF)** and TypeScript (**NodeJs**), ensuring seamless integration between backend and frontend (**Angular/Ionic**).
- **Team Management:** Acted as Lead Developer, coordinating with third-party vendors and internal stakeholders to deliver government-level software solutions.

Freelance | Argentina Fullstack Developer (IoT Specialist)

| 2022

- **IoT Solution Design:** Executed end-to-end development of an IoT state-management system integrating RFID technology, microcontrollers, and MQTT protocols.
- **Tech Stack:** Built the ecosystem using TypeScript, Node.js, Express, C++, Docker, and Nginx.

Argentine Air Force | Argentina R&D Engineer & Team

Lead | 2014 – 2020

- **Radar Signal Processing (2018-2020):** Led R&D efforts for upgrading radar signal processors. Developed embedded systems and algorithms using **Python and C++** for real-time signal analysis (High transferability to AI/ML logic).
 - **IT Operations Management (2014-2020):** Head of Communications and IT Services. Managed a team of 17 professionals, overseeing critical infrastructure (Linux, Cisco) and resource budgeting.
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Workshops & Continuous Learning

BiznagaFest 2025: Advanced AI & MCP Track | Málaga | 2025

- **MCP Servers & Integration:** Practical implementation of Model Context Protocol (MCP) servers with Gisela Torres (Microsoft).
- **Agent Security:** Deep dive into security risks in Agentic AI (exfiltration, backdoors) and securing tool access.
- **AgentOps & Development:** Building diagnostic agents from scratch (Vertex AI SDK + osquery), converting Natural Language to SQL for system monitoring.

Google Cloud: Startup Technical Guide to AI Agents | Google Safety Center | 2025

- *Focus:* End-to-end Agent lifecycle (Prototype to Production). Implementation of **RAG (Retrieval-Augmented Generation)**, multimodal capabilities with Gemini, and **AgentOps** best practices for quality and safety.

Academic Education

Internet of Things (IoT) Specialty | University of Buenos Aires (UBA) | 2020 - 2023

Focus: MQTT, TCP/IP, LoRaWAN, Embedded Systems, Cybersecurity.

Degree in Computer Science | National University of Lanús | 2020 - 2022

Focus: Systems Design, Computational Thinking.

Diploma on Data Science | ITBA | 2020

Focus: Advanced SQL, ETL, Python, Machine Learning/Neural Networks.

Cardano Blockchain Developer | Emurgo | 2021 - 2022

Focus: Haskell, Blockchain fundamentals.

Language Skills

English: Full Professional Proficiency (Daily use in international teams)

Spanish: Native